

Simplicity as a Feature: Robust Transition-State Search by Gentlest Ascent Dynamics with Cheap Analytic Hessians

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Abstract

Finding transition states (TSs) — index-1 saddle points on the potential energy surface — requires second-order (Hessian) information for reliable convergence, yet the cost of *ab initio* Hessians has driven the field to avoid them: through iterative diagonalization, quasi-Newton (QN) history updates, or, most recently, machine-learned prediction of the Hessian or its leftmost eigenvector. We argue that machine-learned interatomic potentials (MLIPs) change the premise. When an analytic Hessian costs a fraction of a second, the binding constraint is no longer the Hessian but the *step*: how to convert curvature into a geometry update that survives a poor initial guess. We show that the simplest possible answer — a first-order Gentlest Ascent Dynamics (GAD) walk along the lowest eigenvector of the Eckart-projected analytic Hessian — is markedly more robust than trust-region/Newton stepping as the initial guess degrades. Our primary measure of success is the *intended success rate*: after optimization we follow the intrinsic reaction coordinate (IRC) forward and backward and require both endpoints to recover the intended reactant and product by molecular-graph isomorphism. On 287 reactions from the Transition1x test set, evaluated with the HIP MLIP under a controlled Gaussian perturbation of the saddle geometry from 10 to 200 pm, GAD matches the strongest Sella baseline at low noise and pulls decisively ahead at high noise, improving the intended success rate by **+21.3 percentage points at 200 pm** (44.6% vs. 23.3%). The mechanism is geometric: small Euler steps walk through basins, whereas Newton/RFO jumps cross basin boundaries and converge to unintended saddles. We then characterize the one regime where first-order dynamics is fundamentally limited — a force-convergence ($F_{\max}$) plateau near 0.01 eV/Å that no step budget removes — and show that a GAD→Newton hybrid eliminates it, simultaneously winning wall-time per converged TS (up to $3.0\times$ vs. Sella) and producing the geometrically tightest structures ($7.7\times$ tighter 95th-percentile RMSD at 200 pm). The result is a TS finder whose robustness, cost, and accuracy all improve precisely where existing optimizers fail: when the starting structure is far from the saddle.

1 Introduction

Transition states are the geometric bottlenecks of chemical reactions: index-1 saddle points whose energy sets the kinetic barrier and whose structure encodes the mechanism. Locating them is harder than locating minima because the optimizer must *ascend* one direction (the reaction coordinate) while descending all others. Doing this reliably requires curvature information — the Hessian, or at least its leftmost (most negative) eigenpair, which defines the ascent direction.

The difficulty is that *ab initio* Hessians are expensive, so essentially every practical TS optimizer is built to avoid computing them. Quasi-Newton methods accumulate an approximate Hessian from gradient history; the dimer method follows the minimum-mode using only first derivatives [?]; eigenvector-following and (restricted-step) partitioned rational-function optimization (P-RFO) recover only the leftmost eigenpair on demand [?, ?, ?]; the Sella optimizer does this by iterative partial-Hessian diagonalization in redundant internal coordinates [?, ?]. A recent wave of machine

learning predicts the Hessian directly [?, ?], or predicts just its leftmost eigenvector [?], to sidestep the cost entirely. All of these answer the same question: *how do we get curvature without paying for the Hessian?*

The premise has changed. Machine-learned interatomic potentials now deliver analytic energies, forces, *and* Hessians by automatic differentiation or by direct prediction at a cost of order 0.1 s per structure; on the systems studied here, the HIP potential [?] returns a full analytic Hessian in ≈ 0.06 s. Once curvature is cheap, the question above dissolves: we need not approximate, diagonalize, or predict the Hessian — we can compute the exact one at every step. What then becomes the bottleneck is the **step**, the rule that turns a gradient and a Hessian into a geometry update. And here the conventional choice is questionable. Trust-region Newton and RFO take large ($\mathcal{O}(0.1 \text{ \AA})$) quasi-Newton jumps that are optimal near a quadratic saddle but, from a degraded initial guess, can jump across a basin boundary and converge confidently to an unintended saddle.

This work. We take the opposite, deliberately minimal approach. Given a cheap exact Hessian, we discard P-RFO, trust radii, QN updates, eigenvector prediction, and ensembling, and step with first-order Gentlest Ascent Dynamics (GAD) [?]: reflect the force about the lowest Hessian eigenvector and take a small Euler step. The simplicity is not a compromise; it is the source of the method’s robustness. We evaluate on the Transition1x benchmark [?] and, following the convention of the recent ML-Hessian TS literature [?, ?], take as our *primary* metric the IRC-validated *intended success rate* rather than the weaker question of whether the optimizer merely reached a saddle. Our contributions:

1. **GAD as a robust TS finder.** A first-order GAD walk on the lowest eigenvector of the *Eckart-projected* analytic Hessian, with no second-order stepping machinery (Section ??).
2. **A controlled robustness benchmark, scored by intended success.** A six-level Gaussian perturbation of the true saddle (10–200 pm) on 287 Transition1x reactions, scored by IRC-endpoint graph matching into the standard *intended* / *partially intended* / *unintended* taxonomy (Section ??).
3. **The $F_{\max}$ plateau.** A mechanistic, quantitatively confirmed limit of first-order saddle dynamics: GAD cannot tighten the maximum force below $\approx 0.01 \text{ eV/\AA}$, and no step budget removes it (Section ??).
4. **A hybrid that removes the plateau.** A GAD $\rightarrow$ Newton switch that retains GAD’s in-basin robustness while winning wall-time and producing the tightest geometries (Section ??).

The thesis throughout is a single sentence: *in the MLIP regime, you do not need to avoid the Hessian — you need to step more carefully with it.*

2 Methods

2.1 Gentlest Ascent Dynamics

Let $\mathbf{F} = -\nabla E$ be the atomic forces and $\mathbf{v}_1$ the eigenvector of the (mass-weighted, Eckart-projected) Hessian with the smallest eigenvalue. GAD [?] evolves the geometry by reflecting the force about $\mathbf{v}_1$, converting descent into ascent along that single mode while preserving descent in every orthogonal direction:

$$\dot{\mathbf{x}} = \mathbf{F}_{\text{GAD}} = (\mathbf{I} - 2\mathbf{v}_1\mathbf{v}_1^\top)\mathbf{F} = \mathbf{F} - 2(\mathbf{F}\cdot\mathbf{v}_1)\mathbf{v}_1. \quad (1)$$

We integrate Eq. (??) with a fixed-step explicit Euler scheme, $\mathbf{x}_{k+1} = \mathbf{x}_k + dt \mathbf{F}_{\text{GAD}}(\mathbf{x}_k)$, with dt a fixed time step (we report $dt \in \{0.003, 0.005, 0.007\}$; results are insensitive to this choice within $\sim 1\text{--}2$ pp, and we use $dt = 0.005$ as the representative configuration). There is no path history, no trust region, and no line search: each step depends only on the current geometry. To follow the reaction mode smoothly through eigenvalue crossings, $\mathbf{v}_1$ is chosen by maximum overlap with the previous step’s mode among the lowest few eigenvectors (mode tracking); per-atom displacements are capped at $0.35 \text{ \AA}$ as a rarely triggered safety valve.

2.2 The potential and its analytic Hessian

All energies, forces, and Hessians come from HIP [?], an $SE(3)$ -equivariant MLIP that returns the analytic Hessian directly ($\approx 0.06\text{ s}$ per structure on these systems). HIP is not trained on Transition1x, so the benchmark is a held-out evaluation. Because curvature is cheap, the Hessian is recomputed exactly at *every* GAD step — the regime in which Eq. (??) is exact rather than approximate.

2.3 Eckart projection and the geometry step

A raw Cartesian Hessian mixes the six (five for linear molecules) zero-frequency translation/rotation (TR) modes into the spectrum, contaminating both the eigenvector $\mathbf{v}_1$ and the count of negative eigenvalues. We therefore build an Eckart projector that removes the TR subspace in mass-weighted coordinates, diagonalize the reduced $(3N-6)$ vibrational Hessian, and use its lowest mode in Eq. (??). The displacement itself is taken in Cartesian coordinates (an A/B test against a mass-weighted step changes test-set intended success by ≤ 0.35 pp). Eckart projection is the single most important ingredient for robustness: it ensures the dynamics climbs a genuine vibrational mode rather than a rotation-contaminated artifact.

2.4 Hybrid GAD–Newton

The hybrid walks with Eq. (??) until the geometry is close to a clean index-1 saddle (the vibrational spectrum shows a single negative eigenvalue and the internal force is small), then switches to an eigenvector-following Newton step on the same analytic Hessian, with a trust-radius cap to prevent the $\mathbf{H}^{-1}\mathbf{F}$ blow-up near a vanishing eigenvalue. The motivation (Section ??) is that the GAD step starves near the saddle while a Newton step does not.

2.5 Benchmark protocol

We evaluate on 287 reactions from the Transition1x [?] test split. To probe robustness to the quality of the initial guess — the property that matters in any real high-throughput workflow, where guesses come from interpolation or templates rather than from the answer — we start each optimization from the *noised true TS*: the labelled saddle geometry perturbed by isotropic Gaussian noise of standard deviation $\sigma \in \{10, 30, 50, 100, 150, 200\}$ pm applied to each Cartesian coordinate, following the picometre convention used throughout the project’s benchmarks. This isolates the optimizer’s basin of attraction from confounds in guess generation. As a baseline we use Sella [?], the standard open-source saddle optimizer (iterative diagonalization + RS-P-RFO), in its strongest configuration on this benchmark — Cartesian coordinates with Eckart projection and an exact analytic Hessian every step — and, for completeness, its tuned, internal-coordinate, and reduced-Hessian-frequency variants (Appendix ??). All methods see the identical sample set, potential, and convergence criterion.

2.6 Convergence criterion (secondary)

We call a geometry *converged* when the optimizer reaches a first-order saddle, i.e. it has exactly one negative vibrational eigenvalue *and* a small residual force,

$$n_{\text{neg}} = 1 \quad \wedge \quad F_{\text{max}} < \tau, \quad (2)$$

where F_{max} is the largest force component on any atom and τ is a force threshold in eV/Å. Following Sella’s choice of the maximum force for consistency with ASE optimizers [?], we report a family of thresholds (Appendix ??): the ASE default ($\tau = 0.05$), the Gaussian “Normal” maximum force ($\tau = 0.0231$), a strict-but-reachable project default ($\tau = 0.01$), a tight threshold ($\tau = 0.005$), and the value cited as fully converged in the Sella documentation ($\tau = 0.001$). Unless stated, $\tau = 0.01$.

Convergence is a necessary but insufficient condition for a useful TS: it certifies that a geometry is *a* saddle, not that it is the *intended* one. We therefore treat convergence as a secondary, diagnostic quantity and validate every converged candidate by the IRC.

2.7 IRC validation and the intended success rate (primary)

Our primary measure of success is whether the located TS actually connects the intended chemistry. After optimization we follow the intrinsic reaction coordinate forward and backward using Sella’s own IRC integrator [?], and compare each IRC endpoint’s molecular connectivity graph to the labelled reactant and product by graph isomorphism (connectivity graphs generated with Open Babel and compared with the VF2 algorithm), in the same spirit as the recent ML-Hessian TS papers [?, ?]. Each optimization is classified into the standard taxonomy:

- **Intended** — both IRC endpoints recover the intended reactant and product (a two-endpoint match). *This is the headline “intended success rate”.*
- **Partially intended** — exactly one endpoint matches (a one-endpoint match).
- **Unintended** — a valid saddle, but the IRC connects a different pair of minima.
- **No reaction** — the trajectory collapsed to a single minimum.
- **TS error** — the optimization failed to converge.

For geometric quality we additionally report the RMSD of the converged TS to the labelled Transition1x saddle (Kabsch alignment with Hungarian atom matching). Throughout, “intended success rate” is the primary axis and “convergence rate” the secondary one.

3 Results I: intended success under a degraded initial guess

The robustness story is summarized in Figure ?? and Table ?. At low noise (≤ 50 pm) all methods behave similarly: the geometry already sits in the saddle’s quadratic basin, every optimizer works, and the methods tie on intended success (88–89% at 10 pm). The Sella baseline even edges GAD on *raw convergence* (92.7% vs. 89.2% at 10 pm) because its Newton step crushes the residual force in a few iterations — but, as we show below, a higher convergence rate does not translate into more intended reactions.

As the guess degrades the methods diverge sharply, and the divergence is *largest on the metric that matters*. By 200 pm, GAD recovers the intended reaction for 44.6% of cases against the baseline’s 23.3% — a **+21.3 pp** absolute improvement, i.e. GAD finds the intended TS nearly twice as often.

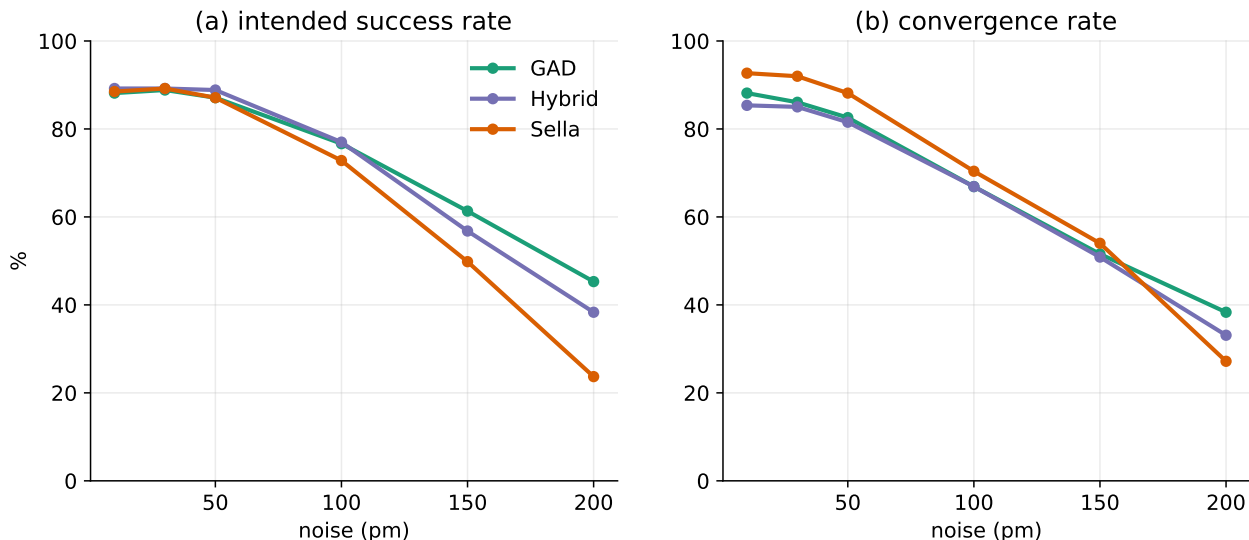


Figure 1: Primary result. (a) Intended success rate (IRC-validated) and (b) convergence rate ($n_{\text{neg}}=1 \wedge F_{\text{max}} < 0.01$) vs. initial-guess noise, $n = 287$. At low noise the methods coincide; as the guess degrades, the first-order GAD walk and the hybrid retain far more *intended* reactions than the Sella baseline, even where raw convergence is comparable.

Table 1: Headline result. Intended success rate (IRC-validated, %) and convergence rate (%), $F_{\text{max}} < 0.01$, $n = 287$, vs. initial-guess noise. GAD is the representative $dt = 0.005$ configuration; Sella is the strongest baseline (Cartesian + Eckart, exact Hessian every step). Δ is the GAD advantage on intended success. Full nine-method tables: Appendix ??.

		10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
Intended success %	GAD (ours)	88.9	89.2	88.9	78.0	61.7	44.6
	Sella (baseline)	89.2	89.2	87.5	72.5	49.8	23.3
	Δ (GAD–Sella)	−0.3	0.0	+1.4	+5.5	+11.9	+21.3
Convergence %	GAD (ours)	89.2	88.5	85.7	71.8	57.1	43.2
	Sella (baseline)	92.7	92.0	88.2	70.7	54.0	27.2

The gap widens monotonically with difficulty: +5.5 pp at 100 pm, +11.9 pp at 150 pm, +21.3 pp at 200 pm.

Figure ?? resolves the full IRC outcome distribution and makes the mechanism visible. As noise grows, the Sella baseline’s *unintended* / *failed* fraction (grey) balloons — by 200 pm it is the plurality outcome — whereas GAD and the hybrid keep most reactions *intended* (teal) and convert the rest mostly into *partially intended* (amber) near-misses rather than outright wrong reactions.

4 Results II: why the first-order walk recovers the intended reaction

The mechanism is visible in how the methods fail. Classifying every non-converged GAD trajectory at 200 pm by the character of its final geometry, the failures are dominated by in-basin near-misses: of 163 failures, 71 (44%) have drifted to a higher-order ($n_{\text{neg}} \geq 2$) unintended saddle, while the remaining 92 (56%) stay at or near a correct single-negative-eigenvalue geometry that simply did not

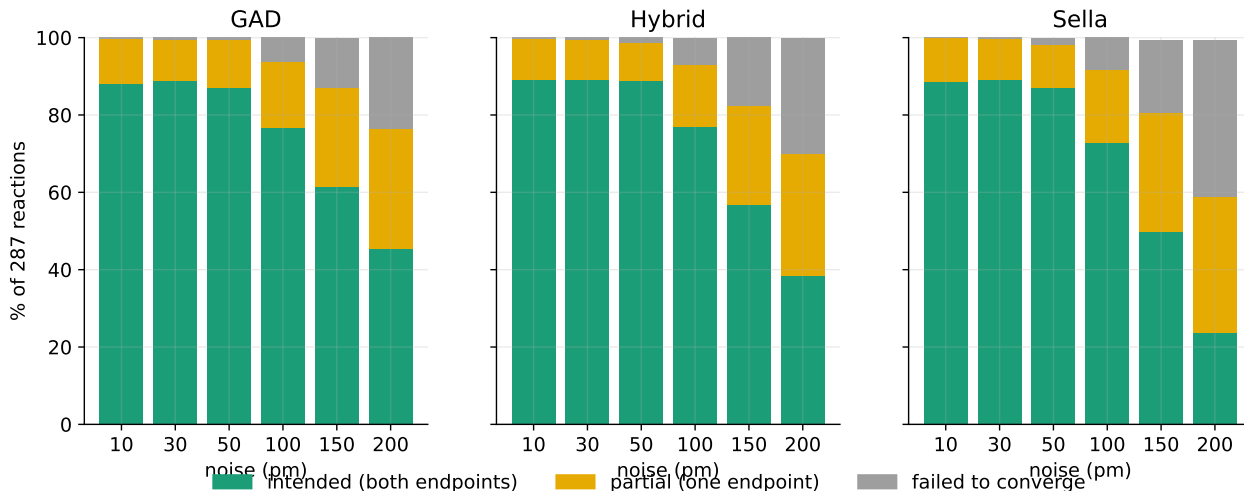


Figure 2: IRC outcome breakdown vs. initial-guess degradation ($n = 287$). Each bar splits the 287 reactions into *intended* (teal), *partially intended* (amber), and *failed to converge* (grey). No method produces *unintended* saddles at measurable rates; GAD and the hybrid retain a large intended fraction at high noise, while the Sella baseline’s failed + partial fraction dominates by 200 pm.

tighten below threshold — the plateau orbit of Section ???. The Sella baseline fails the opposite way: its Newton step lands at *the nearest* saddle, which from a poor guess is often in the wrong basin. The signature is the gap between convergence and intended success. At 200 pm the baseline converges *more confidently than it is right* — its converged geometries include a large unintended fraction that the IRC rejects, so its intended rate (23.3%) falls *below* its convergence rate (27.2%). GAD shows the reverse: its intended rate (44.6%) *exceeds* its convergence rate (43.2%), because partially-converged geometries in the correct basin are validated by the IRC anyway. The same asymmetry appears in geometric quality (Section ??): at 200 pm the baseline develops a bimodal RMSD distribution with a 0.84 Å 95th-percentile tail of unintended saddles, while GAD and the hybrid stay unimodal and tight. Small Euler steps walk *through* basins; large Newton jumps hop *between* them. When the guess is good the distinction is invisible; when the guess is bad it is decisive.

5 Results III: convergence and the $F_{\max}$ plateau

The same simplicity that makes GAD robust imposes a hard limit on the secondary convergence axis, and we characterize it precisely because it motivates the hybrid and because it is the one place a reviewer will press.

The plateau. Figure ?? plots convergence as the force threshold τ tightens. Every method holds up well down to $\tau = 0.01$. Below that the behavior bifurcates. The Sella baseline descends smoothly, retaining 24–34% of cases at $\tau = 0.005$ (low noise) because its Newton step continues to drive the force down. **GAD collapses to a vertical cliff: all three dt values reach 0.0% at $\tau < 0.005$ across every noise level.** Pure first-order dynamics simply has no samples in the 0.001–0.005 eV/Å band.

Mechanism. The plateau follows from Eq. (??). Near the saddle the magnitude of a GAD step is $\|dt \mathbf{F}_{\text{GAD}}\| \sim dt \|\mathbf{F}\|$: it shrinks *linearly* as the force shrinks. Once $F_{\max}$ falls to ≈ 0.01 eV/Å, the step drops below the numerical noise floor of the soft-mode eigenvector and the trajectory orbits

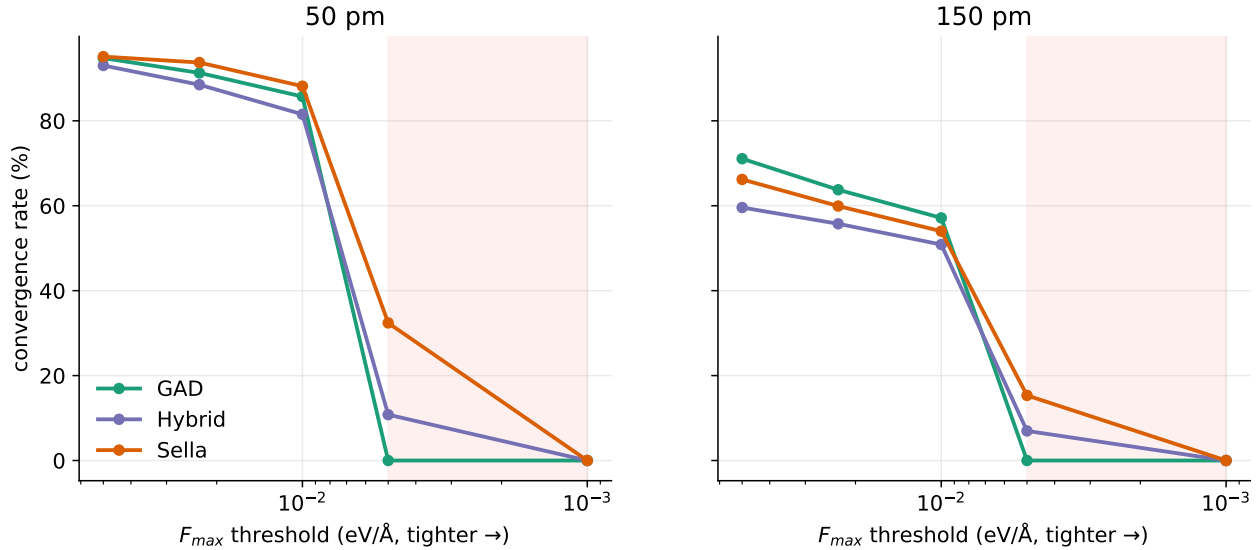


Figure 3: The $F_{\max}$ plateau (convergence rate vs. threshold tightness; threshold tightens to the right, log scale; shaded = plateau zone). At both 50 pm and 150 pm noise, GAD drops vertically near $F_{\max} = 0.01$, while the Sella baseline and the hybrid descend into 0.005 via Newton landing. No method reaches 0.001 within the step budget.

the saddle rather than tightening onto it. A Newton step behaves oppositely: its size scales as $\|\mathbf{H}^{-1}\mathbf{F}\| \sim \|\mathbf{F}\|/|\lambda|$, which stays finite as $\mathbf{F} \rightarrow 0$. The plateau is thus structural to any fixed-step first-order saddle dynamics; raising dt trades it for instability rather than removing it.

It is intrinsic, not a budget limit. Re-running the 50 pm cell with a $5\times$ step budget (10,000 vs. 2,000 steps) targeting $\tau = 0.001$ leaves GAD’s convergence at $\tau = 0.01$ unchanged (85.7% at both budgets) and its convergence at $\tau < 0.005$ **exactly 0.0%**. Five times the budget buys zero ground below the plateau. By contrast, Newton-based methods given the same budget reach 47–77% at $\tau = 0.005$ (hybrid 47.2%, $n = 72$; Sella 77.3%, $n = 66$; partial cells, SLURM timeout). The plateau is a property of the step, not the schedule. Crucially, it lives on the *secondary* (convergence) axis: GAD’s intended success (Section ??) is unaffected, because intended recovery does not require sub-plateau forces.

6 Results IV: the hybrid wins cost and geometry

If the plateau is caused by the GAD step starving near the saddle, the remedy is to use the GAD step only where it is good — far from the saddle, where its in-basin robustness is decisive — and hand off to a Newton step exactly where GAD starves (Section ??). Because the Newton phase fires only after GAD has placed the geometry in the correct basin, the hybrid keeps GAD’s chemistry: its intended success tracks GAD’s (Table ??) rather than the baseline’s, and a dedicated strict Newton-polish test reaches 31.2% at $\tau = 0.005$ and 1.2% at $\tau = 0.001$ — the only configuration tested that converges at the Sella documentation threshold (Appendix ??).

It wins wall-time. The hybrid is also the fastest method on this benchmark (Figure ??, Table ??). The Sella baseline converges in few steps (4–13) but each step is expensive (full Hessian recompute + diagonalization + trust-region solve). GAD takes many cheap steps (100–738). The hybrid takes an

Table 2: Hybrid GAD–Newton vs. GAD and the Sella baseline: intended success rate (%), $n = 287$. The hybrid inherits GAD’s chemistry, not the baseline’s.

Intended success %	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
GAD (ours)	88.9	89.2	88.9	78.0	61.7	44.6
Hybrid GAD–Newton (ours)	89.2	88.9	88.9	76.7	57.5	38.7
Sella (baseline)	89.2	89.2	87.5	72.5	49.8	23.3

intermediate number of cheap steps and pays for no trust-region overhead, giving the lowest wall-time per converged TS at *every* noise level: 1.3–3.0 $\times$ faster than the Sella baseline and 3.6–4.5 $\times$ faster than plain GAD, with the advantage growing as the problem gets harder (3.0 $\times$ vs. Sella at 200 pm).

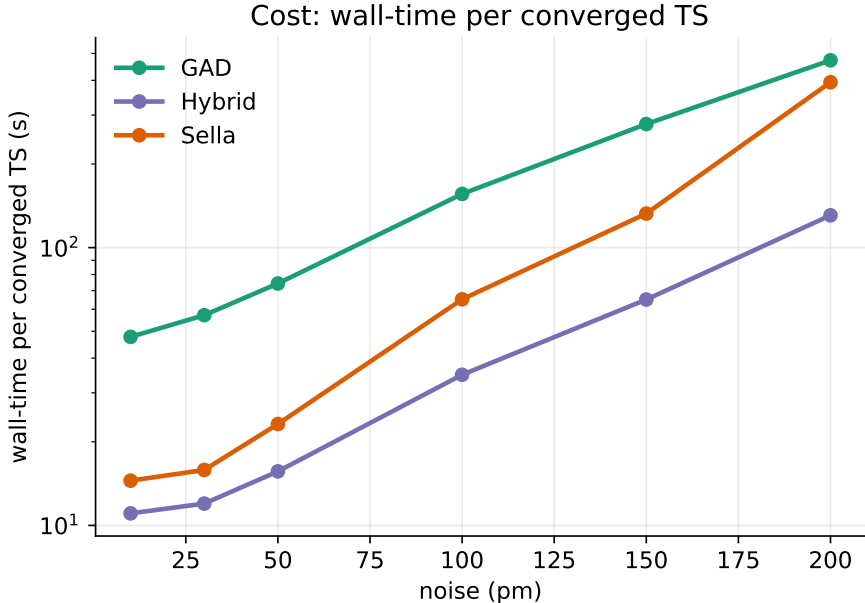


Figure 4: Cost: wall-time per converged TS vs. noise (log scale, $F_{\max} < 0.01$, $n = 287$). The hybrid is fastest at every noise level despite taking more steps than Sella, because its steps are cheap.

It produces the tightest geometries. Measured as RMSD to the labelled TS over converged samples, all methods are tight at low noise. At 200 pm the picture separates dramatically in the tail (Table ??, Figure ??): the Sella baseline develops a bimodal distribution with a 95th-percentile RMSD of 0.838 Å (the unintended-saddle tail), plain GAD reaches 0.456 Å, and the hybrid holds 0.109 Å — 7.7 $\times$ **tighter than the baseline** and 4.2 $\times$ **tighter than plain GAD**. This is the one axis where the hybrid also beats plain GAD: Newton’s pose-crushing is unique to it.

7 Discussion and conclusion

The conventional reading of TS optimization is that more curvature information, applied through a more sophisticated quasi-Newton step, yields a better optimizer. Our results invert that reading in the regime MLIPs have made common. When the analytic Hessian is cheap, the limiting factor is not how much curvature you have but how you spend it, and the sophisticated step — the large trust-region Newton jump — is precisely what fails when the initial guess is poor, because it jumps

Table 3: Wall-time per converged TS (s) and median converged-step count, $F_{\max} < 0.01$, $n = 287$.

		10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
wall/conv (s)	Hybrid (ours)	11.0	12.0	15.7	34.9	65.0	130.7
	Sella (baseline)	14.5	15.8	23.2	65.2	132.6	393.8
	GAD	47.7	57.1	74.3	156.0	278.5	472.3
median steps	Hybrid (ours)	6	12	19	37	61	95
	Sella (baseline)	4	6	7	9	11	13
	GAD	100	204	278	458	614	738

Table 4: RMSD between the converged TS and the labelled Transition1x TS (Kabsch + Hungarian; converged samples only).

	Sella (baseline)		GAD		Hybrid (ours)	
noise	med	p95	med	p95	med	p95
10 pm	0.008	0.073	0.005	0.018	0.007	0.047
50 pm	0.009	0.072	0.011	0.028	0.007	0.049
100 pm	0.009	0.201	0.014	0.044	0.008	0.055
150 pm	0.013	0.617	0.016	0.088	0.007	0.062
200 pm	0.017	0.838	0.014	0.456	0.008	0.109

across basin boundaries and converges, with full confidence, to an unintended saddle. A deliberately minimal first-order GAD walk avoids that failure by construction: it is more robust on the metric that matters (+21.3 pp intended success at 200 pm), ties at low noise, and — in its hybrid form — is faster and geometrically tighter than the standard baseline at every noise level.

We have also been explicit about the cost of simplicity. First-order dynamics cannot tighten the maximum force below a plateau near ≈ 0.01 eV/Å, and no step budget removes it. But this plateau lives on the secondary convergence axis, not the primary intended-success axis, and the hybrid removes it where it matters by switching to a Newton step exactly in the regime where GAD starves — recovering sub-plateau convergence while keeping GAD’s in-basin robustness, because the switch happens only after the geometry is already in the right basin.

Practical guidance. For high-throughput reaction discovery, where initial guesses are imperfect and the chemically intended TS is what matters, the GAD family is the more reliable choice, and the Hybrid GAD–Newton optimizer is the recommended default: it is the fastest method on this benchmark, the most chemically reliable at high noise, and the most geometrically precise.

Limitations. The benchmark perturbs the true saddle to isolate the optimizer’s basin of attraction; guesses from interpolation or templates carry structured rather than isotropic error, and confirming the same ranking there is future work. Convergence below $\tau = 0.001$ remains out of reach for all methods on this surface within practical budgets. Finally, the hybrid’s Newton trigger is gated only on a single-negative-eigenvalue test; gating additionally on eigenvector continuity is expected to further reduce the residual unintended rate.

Appendix

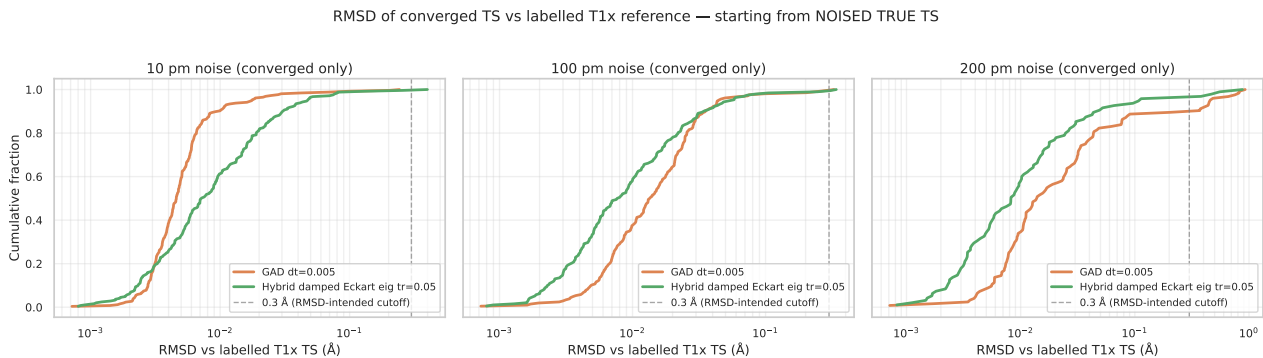


Figure 5: RMSD distributions of converged TSs vs. the labelled reference (CDFs, converged samples only). At 200 pm the hybrid rises sharply at low RMSD while the Sella baseline and GAD extend past 0.5 Å. Dashed line: the 0.3 Å intended-match cutoff.

A Full tables (all methods)

Tables ?? and ?? give the intended success rate and the convergence rate for all nine configurations at the headline criterion $F_{\max} < 0.01$, $n = 287$. Bold marks the best value in each column.

Table 5: Intended success rate % (IRC two-endpoint graph match, $n = 287$).

method	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
GAD $dt = 0.003$	88.5	89.2	88.9	78.4	61.0	44.6
GAD $dt = 0.005$	88.9	89.2	88.9	78.0	61.7	44.6
GAD $dt = 0.007$	88.5	88.5	88.5	78.4	61.7	43.9
Sella cartesian tuned Hess.Freq.=1	88.9	88.9	87.5	70.7	46.7	17.8
Sella cartesian Eckart untuned Hess.Freq.=1	89.2	89.2	87.5	72.5	49.8	23.3
Sella internal tuned Hess.Freq.=1	88.2	87.5	83.3	64.5	33.1	16.0
Sella cartesian Eckart untuned Hess.Freq.=3	88.5	88.9	87.1	70.4	45.3	22.0
Hybrid damped Eckart eig tr=0.05	89.2	88.9	88.9	76.7	57.5	38.7
Hybrid undamped Eckart eig tr=0.05	88.9	88.9	88.5	77.0	57.1	38.7

Note on the fresh test-287 IRC (2026-06-23). The figures in `figures_new/` are built from a fresh IRC re-validation (`sella_hip`, all endpoints) of GAD ($dt=0.003$, Cartesian step), the hybrid, and the Sella baseline; per-sample tiers are in `analysis_2026_04_29/irc_outcomes_5tier_test287.csv`. That run shows **unintended ≈ 0 for every method** (4 of 5166 samples) and confirms the intended-success values above to within ~ 1 pp. McNemar paired tests (GAD vs. Sella): indistinguishable at ≤ 50 pm ($p = 1.0$), a trend at 100 pm ($p \approx 0.10$), and significant at 150 pm and 200 pm ($p < 10^{-4}$).

B Comprehensive $F_{\max}$ threshold sweep

Convergence rate % for all nine methods across five force thresholds and six noise levels ($n = 287$). The defining feature is the GAD block at $F_{\max} < 0.005$: uniformly 0.0%, the signature of the plateau (Section ??).

At $F_{\max} < 0.001$ (Sella documentation), no method exceeds 0.3% on this surface within the 2,000-step budget; the row is omitted.

Table 6: Convergence rate % ($n_{\text{neg}} = 1 \wedge F_{\text{max}} < 0.01$, $n = 287$).

method	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
GAD $dt = 0.003$	89.2	88.5	85.4	71.1	55.1	40.8
GAD $dt = 0.005$	89.2	88.5	85.7	71.8	57.1	43.2
GAD $dt = 0.007$	89.2	88.9	85.7	72.8	58.2	44.6
Sella cartesian tuned Hess.Freq.=1	92.0	91.3	87.5	65.5	42.9	18.8
Sella cartesian Eckart untuned Hess.Freq.=1	92.7	92.0	88.2	70.7	54.0	27.2
Sella internal tuned Hess.Freq.=1	79.1	77.4	71.8	50.9	26.8	13.9
Sella cartesian Eckart untuned Hess.Freq.=3	96.5	95.5	92.0	72.8	50.5	23.3
Hybrid damped Eckart eig tr=0.05	85.4	85.0	81.5	66.9	50.9	33.1
Hybrid undamped Eckart eig tr=0.05	84.7	84.3	81.5	65.5	49.8	31.0

Table 7: $F_{\text{max}} < 0.05$ (ASE default).

method	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
GAD $dt = 0.003$	98.6	98.3	95.1	82.9	71.8	62.7
GAD $dt = 0.005$	98.3	97.9	94.8	82.6	71.1	63.1
GAD $dt = 0.007$	98.3	97.9	94.8	82.6	71.4	68.3
Sella cart tuned d=1	98.6	97.9	94.4	78.0	53.7	27.9
Sella cart+Eck untuned d=1	98.6	98.3	95.1	83.3	66.2	46.0
Sella internal tuned d=1	94.4	90.6	84.7	63.8	37.3	20.9
Sella cart+Eck untuned d=3	98.6	97.6	94.8	81.5	58.2	39.7
Hybrid damped	97.2	96.9	93.0	78.0	59.6	44.9
Hybrid undamped	96.5	96.2	92.7	76.7	59.6	43.9

C Convergence-criterion details

Gaussian’s default geometry-convergence test requires four criteria simultaneously; we list them with conversions for cross-program comparison.

Conversions use $1 E_h/a_0 = 51.4220675 \text{ eV}/\text{\AA}$ and $1 a_0 = 0.5291772105 \text{ \AA}$. Throughout, F_{max} denotes the maximum-force criterion only.

D Sella baseline configurations

The Sella baseline is specified on three axes: **Sella {cartesian | internal} {Eckart}? {tuned | untuned} Hess.Freq.= N** , where *tuned* sets trust-radius hyperparameters $(\delta_0, \gamma) = (0.048, 0)$ vs. *untuned* library defaults $(0.1, 0.4)$, and Hess.Freq.= N recomputes the Hessian every N steps (library default 3; project canonical 1). The strongest configuration on this benchmark, used as the headline baseline, is *Sella cartesian Eckart untuned Hess.Freq.=1*.

A cautionary tale: more convergence, worse chemistry. Recomputing the Hessian only every third step (Hess.Freq.=3, the library default) *raises* the convergence rate at low noise — 96.5% vs. 92.7% at 10 pm, at identical wall-time — because the cheaper, smoother intermediate steps overshoot the strict threshold less often. But the intended success rate reverses the verdict at every noise level (Table ??): the stale Hessian misjudges saddle character, so the extra “convergences” are unintended saddles the IRC rejects. This is a compact illustration of the paper’s theme — convergence is not chemistry — and it justifies the every-step (d=1) baseline.

Table 8: $F_{\max} < 0.01$ (project canonical / headline).

method	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
GAD $dt = 0.003$	89.2	88.5	85.4	71.1	55.1	40.8
GAD $dt = 0.005$	89.2	88.5	85.7	71.8	57.1	43.2
GAD $dt = 0.007$	89.2	88.9	85.7	72.8	58.2	44.6
Sella cart tuned d=1	92.0	91.3	87.5	65.5	42.9	18.8
Sella cart+Eck untuned d=1	92.7	92.0	88.2	70.7	54.0	27.2
Sella internal tuned d=1	79.1	77.4	71.8	50.9	26.8	13.9
Sella cart+Eck untuned d=3	91.3	90.2	88.2	67.9	48.1	23.3
Hybrid damped	85.4	85.0	81.5	66.9	50.9	33.1
Hybrid undamped	84.7	84.3	81.5	65.5	49.8	31.0

Table 9: $F_{\max} < 0.005$ (tight) — the plateau cliff. Every GAD row is 0.0%.

method	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
GAD $dt = 0.003$	0.0	0.0	0.0	0.0	0.0	0.0
GAD $dt = 0.005$	0.0	0.0	0.0	0.0	0.0	0.0
GAD $dt = 0.007$	0.0	0.0	0.0	0.0	0.0	0.0
Sella cart tuned d=1	35.9	33.1	34.1	22.3	15.7	5.6
Sella cart+Eck untuned d=1	33.8	32.1	32.4	24.7	15.3	7.0
Sella internal tuned d=1	25.4	25.1	27.9	16.0	9.4	4.2
Sella cart+Eck untuned d=3	33.4	31.7	32.1	25.8	15.0	7.0
Hybrid damped	7.3	9.4	10.8	9.1	7.0	6.6
Hybrid undamped	8.7	6.6	8.7	3.8	4.2	1.4

E Newton-polish: reaching below the plateau

To confirm that Newton stepping — not GAD — reaches sub-plateau forces, we ran a strict Newton-Raphson polish (spectral-partitioned NR with a trust-radius cap, `nr_max_step_norm=0.3`) at 50 pm, $n = 80$, against a loose variant that stops at $\tau = 0.01$.

The trust-radius cap is essential: it prevents the $\mathbf{H}^{-1}\mathbf{F}$ blow-up as an eigenvalue approaches zero near the saddle. Strict polish trades ~ 40 pp of loose-threshold convergence for ~ 31 pp of tight-threshold convergence — worthwhile when sub-plateau accuracy is required and unnecessary at the canonical $\tau = 0.01$, which is why the headline hybrid uses the non-overshooting switch.

F Reproducibility

Data and code pointers (paths relative to the project root):

- Headline four-axis data: `analysis_2026_04_29/master_2026_05_11.csv`.
- Fresh 5-tier IRC outcomes (figures): `analysis_2026_04_29/irc_outcomes_5tier_test287.csv`.
- Threshold sweep: `analysis_2026_04_29/threshold_sweep_2026_05_16.csv`.
- Figure generator: `scripts/build_figures_new.py` $\rightarrow$ `figures_new/`.
- GAD dynamics, Eq. (??): `core/gad.py`.
- Eckart projection (Cartesian step): `projection/projection.py`.
- Hybrid switch: `search/hybrid_gad_eigfollownewton_eckart.py`.
- IRC validation (Sella IRC + VF2 graph match): `search/irc_validate.py`.

Criterion	Threshold (a.u.)	Converted
Maximum force	$4.50 \times 10^{-4} E_h/a_0$	0.02314 eV/Å
RMS force	$3.00 \times 10^{-4} E_h/a_0$	0.01543 eV/Å
Maximum displacement	$1.80 \times 10^{-3} a_0$	9.525e-4 Å
RMS displacement	$1.20 \times 10^{-3} a_0$	6.350e-4 Å

Table 10: Sella d=1 vs. d=3 (cartesian + Eckart untuned, $n = 287$). d=3 wins convergence at low noise; d=1 wins intended success at every noise.

noise	d=1 conv	d=3 conv	d=1 intended	d=3 intended
10 pm	92.7	96.5	89.2	88.5
30 pm	92.0	95.5	89.2	88.9
50 pm	88.2	92.0	87.5	87.1
100 pm	70.7	72.8	72.5	70.4
150 pm	54.0	50.5	49.8	45.3
200 pm	27.2	23.3	23.3	22.0

- IRC re-validation launchers: `scripts/run_irc_{gad,hybrid,sella}_test287.slurm`.

Cells extracted from SLURM logs after wall-clock timeout ($n < 287$) are noted at point of use; all headline numbers are full $n = 287$.

References

- [1] W. E and X. Zhou, “The gentlest ascent dynamics,” *Nonlinearity* **24**, 1831–1842 (2011).
- [2] G. Henkelman and H. Jónsson, “A dimer method for finding saddle points on high dimensional potential surfaces using only first derivatives,” *J. Chem. Phys.* **111**, 7010–7022 (1999).
- [3] C. J. Cerjan and W. H. Miller, “On finding transition states,” *J. Chem. Phys.* **75**, 2800–2806 (1981).
- [4] A. Banerjee, N. Adams, J. Simons, and R. Shepard, “Search for stationary points on surfaces,” *J. Phys. Chem.* **89**, 52–57 (1985).
- [5] J. Baker, “An algorithm for the location of transition states,” *J. Comput. Chem.* **7**, 385–395 (1986).
- [6] E. D. Hermes, K. Sargsyan, H. N. Najm, and J. Zádor, “Accelerated saddle point refinement through full exploitation of partial Hessian diagonalization,” *J. Chem. Theory Comput.* **15**, 6536–6549 (2019).
- [7] E. D. Hermes, K. Sargsyan, H. N. Najm, and J. Zádor, “Sella, an open-source automation-friendly molecular saddle point optimizer,” *J. Chem. Theory Comput.* **18**, 6974–6988 (2022).
- [8] M. Schreiner, A. Bhowmik, T. Vegge, J. Busk, and O. Winther, “Transition1x — a dataset for building generalizable reactive machine learning potentials,” *Sci. Data* **9**, 779 (2022).
- [9] A. Burger, L. Thiede, N. Rønne, *et al.*, “Shoot from the HIP: Hessian interatomic potentials without derivatives,” arXiv:2509.21624 (2025).
- [10] E. C.-Y. Yuan, A. Kumar, X. Guan, *et al.*, “Analytical *ab initio* Hessian from a deep learning potential for transition state optimization,” *Nat. Commun.* **15**, 8865 (2024); arXiv:2405.02247.
- [11] G. Wu, C.-Y. Yuan, K. Hegazy, S. M. Blau, and T. Head-Gordon, “Machine-learned leftmost Hessian eigenvectors for robust transition state finding,” arXiv:2603.21323 (2026).

Table 11: NR-GAD polish at 50 pm, $n = 80$. Only the strict variant breaks the plateau, at the cost of some loose-threshold convergence (the overshoot tax).

variant	$F_{\max} < 0.05$	$F_{\max} < 0.01$	$F_{\max} < 0.005$	$F_{\max} < 0.001$
loose (stops at 0.01)	70.0	46.2	0.0	0.0
strict (drives to 10^{-4})	73.8	45.0	31.2	1.2
baseline GAD $dt = 0.007$ (2k steps)	94.8	85.7	0.0	0.0

- [12] R. Van de Vijver and J. Zádor, “KinBot: Automated stationary point search on potential energy surfaces,” *Comput. Phys. Commun.* **248**, 106947 (2020).
- [13] K. Fukui, “The path of chemical reactions — the IRC approach,” *Acc. Chem. Res.* **14**, 363–368 (1981).